E-COMMERCE SALES DATASET

PROJECT OBJECTIVE :

The main objective of this Ecommerce Sales Analysis project is to clean, integrate, and analyze two years of ecommerce transaction data from October 2023 to 2024. The analysis aims to identify which products, customers, stores, and regions are driving the maximum revenue for the business. By using Excel tools like VLOOKUP, PivotTables, Slicers, and Charts, an interactive and dynamic sales dashboard is created. This dashboard will help the management team to understand sales performance, product trends, customer buying behaviour, discount impact, and revenue patterns. Finally, the project also focuses on forecasting future sales trends to support effective sales and market.

DATASET DESCRIPTION :

The E-Commerce Sales Dataset is a comprehensive collection of online retail transaction data covering a period of two years from October 2023 to October 2024. The dataset contains 2000+ transaction records with detailed information about customers, products, orders, pricing, and location.The dataset is organized into multiple related tables (dimensions) such as Sales Fact Table, Product Dimension, Customer Dimension, and Store Dimension. The Sales Fact table is the main table which contains transactional details like Sales ID, Customer ID, Product ID, Store ID, Quantity, Unit Price, Discount, and Total Sales Amount.This dataset can be effectively used to analyze sales performance, product-wise trends, customer buying behavior, regional performance, discount impact, and overall revenue patterns. It also helps in identifying the best-selling categories, high-value customers, profitable regions, and payment preferences. The data is suitable for creating PivotTables, Charts, KPIs, and an interactive Excel Dashboard for business decision-making and future sales forecasting.

Key Columns Included:

Month / Order Date: Date of order Placed Sales

Unique ID for each sales transaction :Date of order Placed Sales

Customer ID: Unique ID for each customer

Product ID / Category / Brand: Product details

Quantity & Unit Price: Quantity sold and price per unit

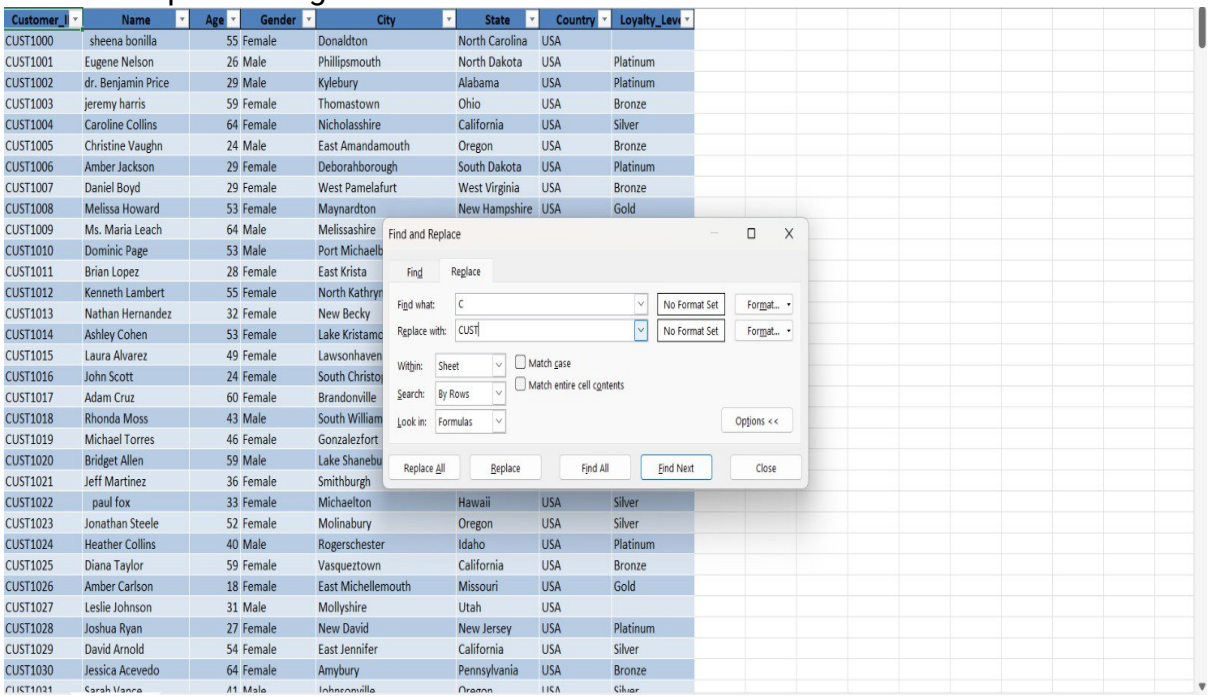
Discount & Total Sales: Discount offered and final sales value

Payment Method / Loyalty: Location and customer details

COLUMN DESCRIPTION :

Column Name	Description
Month / Order Date	Date when the order was placed by the customer
SalesID	Unique transaction ID for each sale (Primary Key
CustomerID	Unique ID to identify each customer
ProductID	Unique ID for each product
ProductCategory	Category of the product

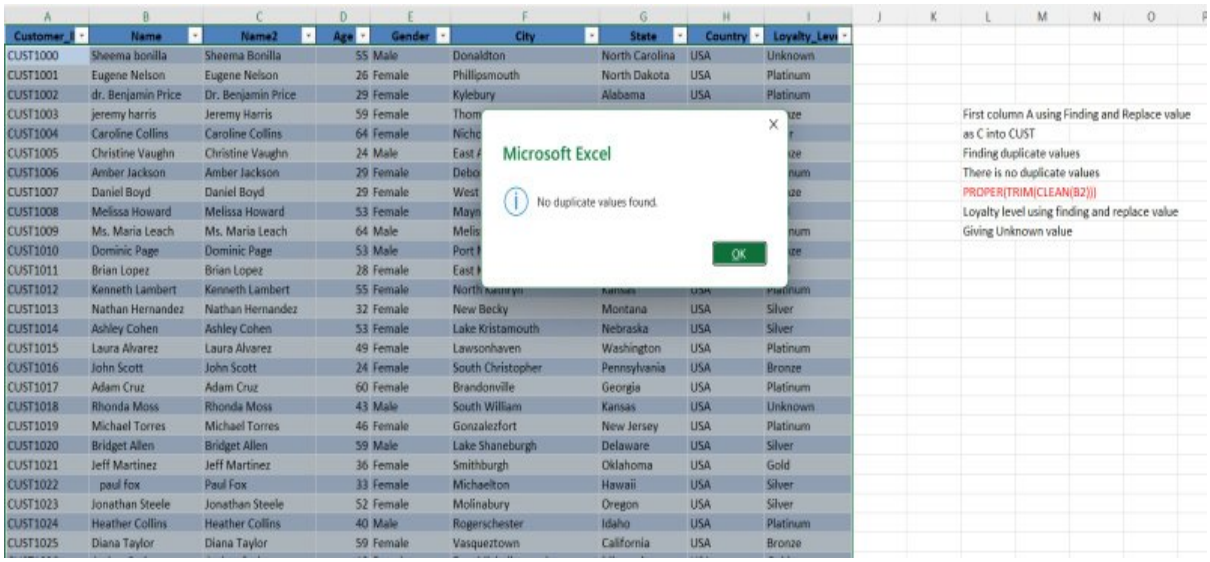
Find and Replace using C to CUST



Customer_I	Name	Age	Gender	City	State	Country	Loyalty_Levi
CUST1000	sheena bonilla	55	Female	Donaldton	North Carolina	USA	
CUST1001	Eugene Nelson	26	Male	Phillipsmouth	North Dakota	USA	Platinum
CUST1002	dr. Benjamin Price	29	Male	Kylebury	Alabama	USA	Platinum
CUST1003	jeremy harris	59	Female	Thomastown	Ohio	USA	Bronze
CUST1004	Caroline Collins	64	Female	Nicholasshire	California	USA	Silver
CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA	Bronze
CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	USA	Platinum
CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA	Bronze
CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire	USA	Gold
CUST1009	Ms. Maria Leach	64	Male	Melissashire			
CUST1010	Dominic Page	53	Male	Port Michaelb			
CUST1011	Brian Lopez	28	Female	East Krista			
CUST1012	Kenneth Lambert	55	Female	North Kathryn			
CUST1013	Nathan Hernandez	32	Female	New Becky			
CUST1014	Ashley Cohen	53	Female	Lake Kristamc			
CUST1015	Laura Alvarez	49	Female	Lawsonhaven			
CUST1016	John Scott	24	Female	South Christo			
CUST1017	Adam Cruz	60	Female	Brandonville			
CUST1018	Rhonda Moss	43	Male	South William			
CUST1019	Michael Torres	46	Female	Gonzalezfort			
CUST1020	Bridget Allen	59	Male	Lake Shanebu			
CUST1021	Jeff Martinez	36	Female	Smithburgh			
CUST1022	paul fox	33	Female	Michaelton	Hawaii	USA	Silver
CUST1023	Jonathan Steele	52	Female	Molinabury	Oregon	USA	Silver
CUST1024	Heather Collins	40	Male	Rogerschester	Idaho	USA	Platinum
CUST1025	Diana Taylor	59	Female	Vasqueztown	California	USA	Bronze
CUST1026	Amber Carlson	18	Female	East Michellemouth	Missouri	USA	Gold
CUST1027	Leslie Johnson	31	Male	Mollyshire	Utah	USA	
CUST1028	Joshua Ryan	27	Female	New David	New Jersey	USA	Platinum
CUST1029	David Arnold	54	Female	East Jennifer	California	USA	Silver
CUST1030	Jessica Acevedo	64	Female	Amybury	Pennsylvania	USA	Bronze
CUST1031	Sarah Vance	41	Male	Johnsonville	Oregon	USA	Silver



Changing the name column correctly



Finding duplicate values

Changing name
PROPER(TRIM(CLEAN(B2)))

Loyalty Level using finding and replace value Giving Unknown value



Product_ID	Product_Name	Category	Sub_Category	Brand	Cost	Stock	Cost	Stock
PROD0	Brother Laser Printer	Electronics	Computer	Hunt Inc	112.42	416	112.42	416
PROD1	Men's Graphic T-Shirt	Clothing	Tops	Terry-Moore	342.56	181	342.56	181
PROD2	Women's Casual Hoodie	Clothing	Outerwear	Sims Inc		352	254.0436	352
PROD3	Men's Running Jacket	Clothing	Outerwear	Pinedo Ltd	364.67	452	364.67	452
PROD4	Women's Yoga Pants	Clothing	Bottoms	Gonzales-Davis	306.46	487	306.46	487
PROD5	Men's Leather Belt	Clothing	Accessories	Baldwin, Garza and Gonzalez	261	59	261	59
PROD6	Herbal Face Moisturizer	Beauty	Skincare	Davidson PLC	312.8	350	312.8	350
PROD7	LED Table Lamp	Home	Lighting	Harrell LLC	218.86	170	218.86	170
PROD8	Wireless Bluetooth Earbuds	Electronics	Audio	Perez-Johnston	439.07	142	439.07	142
PROD9	Adjustable Dumbbell Set	Sports	Fitness	Hill PLC	495.72	323	495.72	323
PROD10	Ceramic Kitchen Bowl Set	Home	Kitchen	Hamilton Group	298.86	308	298.86	308
PROD11	HD Smart TV Stick	Electronics	Accessories	Gibson, Hobbs and Weaver	449.4		449.4	273.4948
PROD12	Hair Strengthening Shampoo	Beauty	Haircare	Rice LLC	115.82	460	115.82	460
PROD13	USB-C Fast Charger	Electronics	Accessories	Dawson-Perkins	242.23	453	242.23	453
PROD14	Decorative Wall Mirror	Home	Decor	Luna, Anderson and Bowman	48.01	439	48.01	439
PROD15	4K Home Projector	Electronics	Appliances	Thomas LLC	465.34	301	465.34	301
PROD16	Beard Grooming Kit	Beauty	Grooming	Erickson-Green	337.42	256	337.42	256
PROD17	Soccer Training Cone Set	Sports	Accessories	Ford, Murphy and Wilson	144.53	367	144.53	367
PROD18	Women's Denim Jeans	Clothing	Bottoms	Guerra-Hendrix		343	254.0436	343
PROD19	Knee Support Brace	Sports	Equipment	May-Reese	21.34	65	21.34	65
PROD20	Outdoor Sports Water Bottle	Sports	Accessories	Soto-Ellis	131.51		131.51	273.4948
PROD21	Multi-Purpose Storage Box	Home	Storage	Gardner-Montes	123.46	76	123.46	76
PROD22	Carbon Fiber Tennis Racket	Sports	Equipment	Flores, Pineda and Rodriguez	484.62	432	484.62	432
PROD23	Men's Formal Shirt	Clothing	Tops	Moon, Mora and Santos		360	254.0436	360
PROD24	Foldable Camping Chair	Sports	Outdoor	Cook Ltd	166.06	340	166.06	340

Removing duplicate values

Next Product ID column using finding and replace

Using Formulas

For finding missing values in Cost

IF(F2="",AVERAGE($F$2:$F$200),F2)

For finding missing values in Stock

IF(G2="",AVERAGE($F$2:$F$200),G2)

(F2Loyalty Level use finding and Replace to Unknown



Removing duplicate values

Formula

For finding missing values in cost

IF (F2="" AVERAGE($F$2: $F$200), F2)

For finding missing values in stock

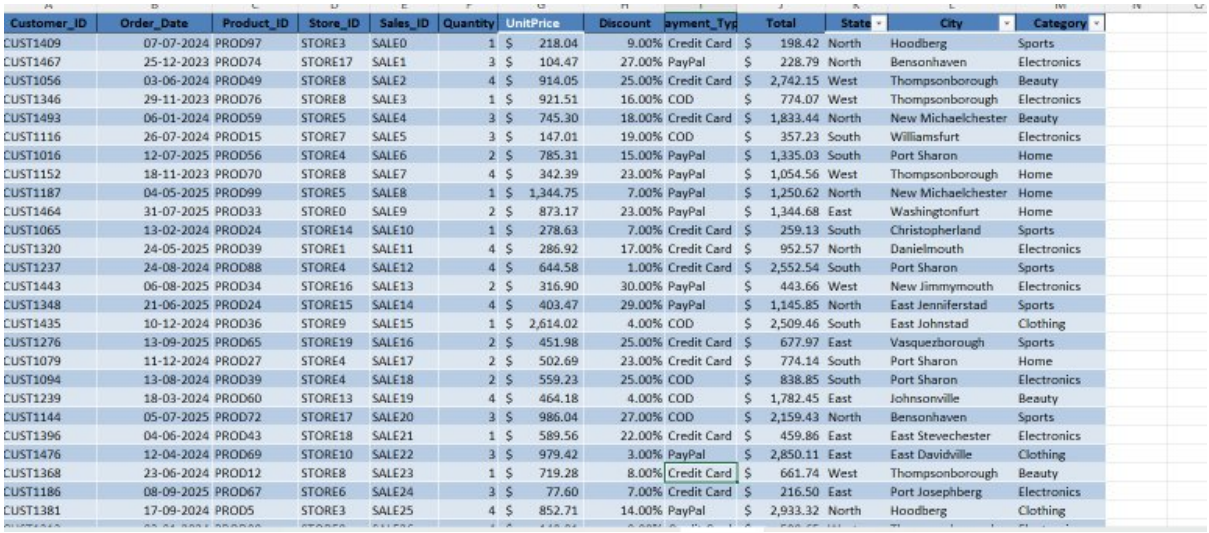
IF (G2="" AVERAGE($F$2: $F$200), G2)

Adding XLOOKUP in final sheet

=XLOOKUP(B2, Customer_Dim!A:A, Customer_Dim!G:G)

=XLOOKUP(E2, Store_Dim!A:A, Store_Dim!D:D, "Not Found")

=XLOOKUP(D2, Product_Dim!A:A, Product_Dim!C:C, "Not Found")



Customer_ID	Order_Date	Product_ID	Store_ID	Sales_ID	Quantity	UnitPrice	Discount	ayment_Typ	Total	State	City	Category
CUST1409	07-07-2024	PROD97	STORE3	SALE0	1	$ 218.04	9.00%	Credit Card	$ 198.42	North	Hoodberg	Sports
CUST1467	25-12-2023	PROD74	STORE17	SALE1	3	$ 104.47	27.00%	PayPal	$ 228.79	North	Bensonhaven	Electronics
CUST1056	03-06-2024	PROD49	STORE8	SALE2	4	$ 914.05	25.00%	Credit Card	$ 2,742.15	West	Thompsonborough	Beauty
CUST1346	29-11-2023	PROD76	STORE8	SALE3	1	$ 921.51	16.00%	COD	$ 774.07	West	Thompsonborough	Electronics
CUST1493	06-01-2024	PROD59	STORE5	SALE4	3	$ 745.30	18.00%	Credit Card	$ 1,833.44	North	New Michaelchester	Beauty
CUST1116	26-07-2024	PROD15	STORE7	SALE5	3	$ 147.01	19.00%	COD	$ 357.23	South	Williamsfurt	Electronics
CUST1016	12-07-2025	PROD56	STORE4	SALE6	2	$ 785.31	15.00%	PayPal	$ 1,335.03	South	Port Sharon	Home
CUST1152	18-11-2023	PROD70	STORE8	SALE7	4	$ 342.39	23.00%	PayPal	$ 1,054.56	West	Thompsonborough	Home
CUST1187	04-05-2025	PROD99	STORE5	SALE8	1	$ 1,344.75	7.00%	PayPal	$ 1,250.62	North	New Michaelchester	Home
CUST1464	31-07-2025	PROD33	STORE0	SALE9	2	$ 873.17	23.00%	PayPal	$ 1,344.68	East	Washingtonfurt	Home
CUST1065	13-02-2024	PROD24	STORE14	SALE10	1	$ 278.63	7.00%	Credit Card	$ 259.13	South	Christopherland	Sports
CUST1320	24-05-2025	PROD39	STORE1	SALE11	4	$ 286.92	17.00%	Credit Card	$ 952.57	North	Danielmouth	Electronics
CUST1237	24-08-2024	PROD88	STORE4	SALE12	4	$ 644.58	1.00%	Credit Card	$ 2,552.54	South	Port Sharon	Sports
CUST1443	06-08-2025	PROD34	STORE16	SALE13	2	$ 316.90	30.00%	PayPal	$ 443.66	West	New Jimmymouth	Electronics
CUST1348	21-06-2025	PROD24	STORE15	SALE14	4	$ 403.47	29.00%	PayPal	$ 1,145.85	North	East Jenniferstad	Sports
CUST1435	10-12-2024	PROD36	STORE9	SALE15	1	$ 2,614.02	4.00%	COD	$ 2,509.46	South	East Johnstad	Clothing
CUST1276	13-09-2025	PROD65	STORE19	SALE16	2	$ 451.98	25.00%	Credit Card	$ 677.97	East	Vasquezborough	Sports
CUST1079	11-12-2024	PROD27	STORE4	SALE17	2	$ 502.69	23.00%	Credit Card	$ 774.14	South	Port Sharon	Home
CUST1094	13-08-2024	PROD39	STORE4	SALE18	2	$ 559.23	25.00%	COD	$ 838.85	South	Port Sharon	Electronics
CUST1239	18-03-2024	PROD60	STORE13	SALE19	4	$ 464.18	4.00%	COD	$ 1,782.45	East	Johnsonville	Beauty
CUST1144	05-07-2025	PROD72	STORE17	SALE20	3	$ 986.04	27.00%	COD	$ 2,159.43	North	Bensonhaven	Sports
CUST1396	04-06-2024	PROD43	STORE18	SALE21	1	$ 589.56	22.00%	Credit Card	$ 459.86	East	East Stevechester	Electronics
CUST1476	12-04-2024	PROD69	STORE10	SALE22	3	$ 979.42	3.00%	PayPal	$ 2,850.11	East	East Davidville	Clothing
CUST1368	23-06-2024	PROD12	STORE8	SALE23	1	$ 719.28	8.00%	Credit Card	$ 661.74	West	Thompsonborough	Beauty
CUST1186	08-09-2025	PROD67	STORE6	SALE24	3	$ 77.60	7.00%	Credit Card	$ 216.50	East	Port Josephberg	Electronics
CUST1381	17-09-2024	PROD5	STORE3	SALE25	4	$ 852.71	14.00%	PayPal	$ 2,933.32	North	Hoodberg	Clothing

Adding Extra columns in final sheet

- State
- Category
- City

Statistics Description Toolpak

	Quantity
Mean	2.446894
Standard Error	0.049923
Median	2
Mode	2
Standard Deviation	1.115195
Sample Variance	1.24366
Kurtosis	-1.34424
Skewness	0.090305
Range	3
Minimum	1
Maximum	4
Sum	1221
Count	499
Largest(1)	4
Smallest(1)	1
Confidence Level(95.0%)	0.098086



Dashboard

Insights

1. Descriptive Analysis - What Happened?
   - Which month had the highest and lowest sales trend?
   - What is the sales comparison between Loyalty Members?
   - Which payment method (Cash, Credit Card, UPI) is used the most?
2. Diagnostic Analysis - Why Did It Happen?
   - Why are sales high even without discounts in some categories?
   - Which category has the highest product returns and why?
   - Why is the performance of one region lower than others?
3. Predictive Analysis - What Will Happen?
   - Which products will have high demand in the next festival season?
   - Can we predict which customers are unlikely to purchase again?
   - What will be the estimated total revenue for the next 6 months?
4. Prescriptive Analysis - What Should We Do?
   - Which products should we restock more to avoid stock-outs?
   - Where should we increase prices and where should we offer more discounts?
   - In which region should the company open its next new store?

Conclusion

In conclusion, the Ecommerce Sales Analysis and Dashboard project successfully analyzes two years of transactional data from October 2023 to 2024 to provide

valuable business insights. Through data cleaning, integration using VLOOKUP, and advanced analysis using PivotTables, Slicers, Timelines, and various charts, an interactive and user-friendly Excel dashboard was created.The dashboard reveals that certain product categories, such as Electronics and Fashion, along with specific regions like North and East, are the major drivers of revenue. It also highlights the impact of discounts, the importance of loyalty customers, and monthly sales trends. The KPIs like Total Sales, Average Sales per Order, and Total Orders provide a quick overview of overall business performance.The forecast feature helps in predicting future sales trends, which supports effective inventory planning and marketing strategies. Overall, this dashboard serves as a powerful decision-making tool for the management to improve sales performance, optimize discount strategies, retain loyal customers, and achieve sustainable business growth in the competitive ecommerce market.